

International Conference on Machine Learning and Data Engineering

Detection of Fake News by Convolutional Neural Networks and Recurrent Neural Networks

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Abstract

The proliferation of fake news on digital platforms poses significant threats to public trust, democracy, and social stability. In this study, a hybrid model by agglomerating the two different neural networks has been introduced for detecting fake news accurately. CNN captures local textual patterns, while the RNN models temporal dependencies for enhanced context understanding. Key data preprocessing techniques such as HTML tag removal, URL elimination, stopword removal, and the use of GloVe word embeddings were applied to ensure data integrity. The proposed model achieved high performance, with a testing accuracy of 91.80%, an F1-score of 0.91, and an Area Under the Curve (AUC) of 0.97, demonstrating its robustness in differentiating news. These results suggest that the hybrid architecture effectively leverages both local and sequential features for improved fake news classification.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Fake news, CNN, RNN, Hybrid models

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1. Introduction

The increase of fake news in an age of digital would become one of the major concerns while attracting researchers, experts, and the public. Also, it has been observed that misinformation as well as disinformation has been well displayed as there has steady increase in the widespread use of online channels and social media platforms. This led to extensive research for understanding the origins, effects, and dissemination patterns. During early research, identifying fake news has been primarily focused on while understanding its key characteristics. For instance, Pennycook and Rand [1] highlighted the urgent need for effective interventions, demonstrating the detrimental impact of misinformation on public perception and the democratic process. during the 2016 US presidential election, Allcott and Gentzkow [2] conducted a comprehensive analysis of fake news which revealed many factors that spread and influence.

For exploring the different psychological and health impacts of fake news, many researchers have explored different things which have been explained in subsequent paragraphs. Vosoughi et al. [3] investigated that fake news has been spreading faster and reaching a wider audience for real news on social media which suggests that novelty and emotional appeal play pivotal roles in applying in different applications Lewandowsky et al. [4] highlighted the influence of cognitive biases, such as confirmation and truth illusion, on the susceptibility of people to misinformation. Many computational methods are available and there has been an increased focus in this area. Natural language processing (NLP) is one of the emerging fields for analyzing contextual meanings and studying the identical patterns that display misinformation [5,6]. In the influencer role in communities, analyzing the network has been employed for examining the spread of fake news on social media [7,8].

Artificial intelligence (AI) [9,10]. algorithms play an important role in developing fake news. Some of algorithms such as Decision Trees (DT) [11], Random Forest (RF) [12], Support Vector Machines (SVM) [13], Neural Networks (NN) [14], and Deep Neural Networks (DNN) [15] has been popularly applied for classifying the fake texts from real texts. From the above studies, it has been shown that significant challenges are faced while detecting fake news. The fake news is identified through behaviour of spreaders and their association with unusual events[16]. Moreover, the algorithms are trained in this manner so that the performance does not get affected. Depending on a few of the uncertain factors, such as credibility and framing, the article can be framed as a true article. Ethical considerations have been one of the potential applications of existing texts where the voices are suppressed in text form [17]. In addition to that, easy attacks can happen by considering the hostile nature and exploiting the malicious actors [18,19].

1.1. Problem Statement

The increase in fake news has posed significant threats to democratic styles, the trust of the public, and social influence. Several existing detection methods have limitations in terms of reliability and practicality. Major challenge faced is the scarcity of properly labeled datasets by developing adaptive generalization techniques. In the current detection systems, some of the limitations have been observed by misinterpreting some of the textual facts which leads to false classifications. By considering the above-related issues, modern technologies can verify the authenticity of news by exposing the contents that are media based., and also enhancing the safeguarding of public discourse by fake news announcements.

2. Proposed Methodology

In this section, the proposed work has been mentioned by following three processes, Data collection, Data preprocessing, and Data analysis which has been described below in subsections.

2.1. Data Collection

Qualitativeness of data used which affect the capabilities of the model that determines the good performance in the model. The file for this research is the "SherLockFakenewsNetOriginal.csv" file obtained from the Kaggle platform, which is widely used as a platform that supports the creation of information and competitive technology. The file "SherLockFakenewsNetOriginal.csv" is a collection of news divided into two categories: real news, fake news, and other online contests. It has many features: For example, scrolling/tiling news headlines is important for training

models to detect fake news., national archives, web access, and fourth, provides commercial information through collaboration. In the case of datasets, this selection is determined by the researcher's goals, data availability, relevance of the data to the problem area, and ethical considerations regarding data security and data use policies.

2.2. Data Preprocessing(DP)

- DP serves as critical stage in machine learning pipeline where during this stage the data is prepared for training and evaluation of the models. During this phase, a number of approaches, such as cleaning, preparation and enrichment of data, are applied to guarantee the data quality that is equal to the requirements for the analysis purposes. For this research project, the "SherLockFakenewsNetOriginal.csv" dataset, obtained from Kaggle, underwent thorough preprocessing to address issues such as missing values, text formatting, and noise reduction. The following steps outline the data preprocessing workflow:

Loading the Dataset: Data was read via the `read_csv()` function and stored inside a Pandas DataFrame. Our dataset is comprised of the news items which are assigned with categories of real or fake news, having features of text and credibility label.

- Sampling the Data: Large dataset to be processed (216 682 rows) Random sample is generated for accelerating processing and representativeness keeps in mind (50% of sample is selected which is a subset of the original data). This goal was accomplished using the `sample()` in Pandas as a tool.
- Handling Missing Values: To deal with the missing values in this data, `fillna()` function has replaced 0 in column 'reliable' which represents expected disqualification of the entry As well, other columns which might be 'NA' or empty strings were replaced with NULL (' ') so the internal data consistency is sustained.
 - Text Preprocessing: When text information is used, the model is may be affected badly by the noise and noise which make it to perform below the required standards.
 - HTML Tag Removal: The BeautifulSoup library was applied to clean HTML tags from the news headlines so that their textual content could be retained while the meaning is passed on.
 - Removing Square Brackets: Or regular expressions were used to delete text which was between square brackets, in which probably annotations are kept or metadata is mentioned.
 - Removing URLs: The website was another regular expression being used to remove URL from text, because they are frequently contain excess data and are not a semantic contribution.
 - Removing Stopwords: These are the words with stop words, they are the frequent words with no much semantic meaning, mostly (e. g. , 's', ' what' , ' stopwords from the NLTK library were removed from the text.
 - Noise Reduction: At the end finally, to cleanse the news headlines the custom function `denoise_text()` was used which sequentially applies the subsequent set of preprocessing steps combining them effectively to reduce the noise.

By performing comprehensive data preprocessing, the raw dataset was transformed into a clean and standardized format which is ready for subsequent stages of the ML pipeline.

2.3. Data Analysis

A word cloud is an visually appealing way to display text data, where each word's size corresponds, to how it appears in the text. In word cloud visualization words are arranged in an aesthetically pleasing manner with common words displayed larger and more prominently while less common words are smaller. Word clouds are commonly used for exploring data to understand the significant words in a dataset. By representing word frequencies word clouds offer a to understand overview of the text content enabling users to quickly spot patterns, themes, and trends. In Figure 1 the word cloud illustrates the word frequencies in text whereas in Figure 2 it represents the word frequencies, in text.

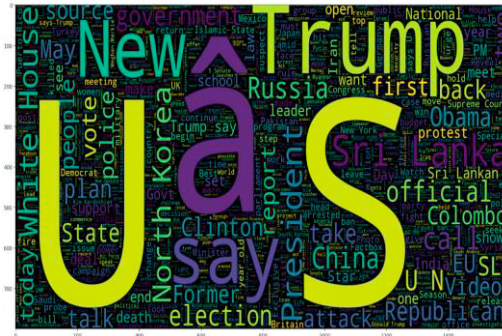


Fig. 1. Illustration of word frequency in form of text

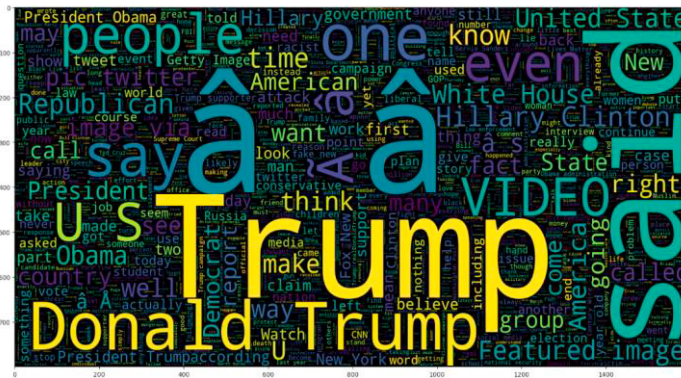


Fig. 2. Frequency of word

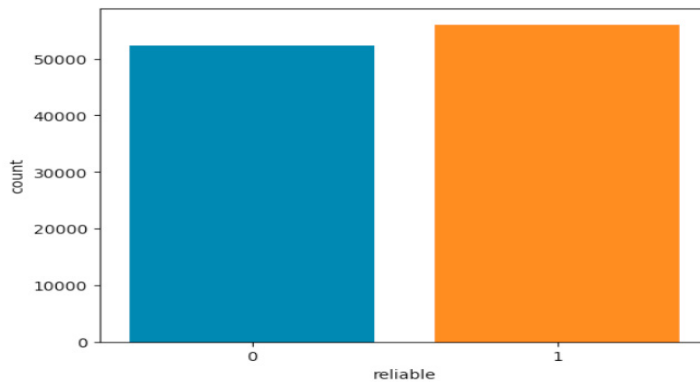


Fig. 3. Summary of histograms for fake and real data

Fig. 3 presents a histogram which visualizes the distribution. In this Fig., the binary labels are assigned during the preprocessing stage where 0 (fake news) and 1 (real news). The x-axis of the histogram displays these two categories. Also, a balanced distribution is ideal for classification tasks, as it ensures the model does not become biased toward one class.

If the figure shows a significant imbalance (e.g., more real news than fake), it implies that the model might become skewed towards predicting the majority class (real news) which resulted in a higher likelihood of missing fake news. The values in the histogram were computed from the sampled dataset, which contained a subset (50%) of the original 216,682 rows. The histogram accurately reflects the proportion of types of news articles in this sample. By using this visualization technique, the sample is used for representing the original dataset which allows for training in the development of the models.

3. Feature Engineering

Feature engineering stands as the crucial feature responsible for providing machine learning models increased representational power by changing the raw data to well-designed features that exposures truthful patterns and associations. When it comes to language processing, feature engineering process is expressed for the purpose of figuring out the significant representations of textual data in order to enhance the prediction.

- **Dataset Spilting:** Data is then fragmented into the trained and tested sets by applying 'train_test_split' from 'scikit-learn' as a part of the Python library. This portion shows that that model is evaluated on data which is not earlier seen by the model to estimate its generalization ability.
- **Tokenization and Padding:** Training set and test set news headlines were tokenized using the Tokenizer class from the Keras library and transformed into Naive-1 sequences of integers. Tokens signify word/letter level and therefore the transformation from text condition to the numerical format that can be utilised by machine learning methods. Moreover, padding was used, which made all sequences regardless of their original length, uniform, determined by the maxlen parameter.
- **Word Embeddings:** The word embeddings are used to capture semantic links that exist between given words by assigning them coordinates in high-dimensional vectors that are never ending. In this project, we employed pre-trained embeddings (GloVe, Global Vectors for Word Representation model) to initialize the embedding matrix. The GloVe embeddings are taken from the '\$GLO=\$VE-', twitter. 27B. 100d. "emobdd.txt" file that incorporates pre-trained Word2Vec vectors trained on Twitter data, characterized by 100-dimensional embeddings.
- **Embedding Matrix Initialization:** The embedding matrix was initialized as well as based on pre-trained word embeddings, consequently, each row is assigned to a word index in the vocabularies of the tokenizer. Compared with the condition when words are absent from the pre-trained embeddings, random values are generated that are drawn from normal distrubution having mean and standard deviation.
- **Integration with Tokenizer:** The embedding matrix was integrated with the tokenizer to ensure that words in the vocabulary are mapped to their corresponding embedding vectors during model training. This integration would improve in capturing the semantic similarities from text data.

3.1 CNN Implementation for Fake News Detection

In the proposed model, CNN architecture has been employed as part of the hybrid CNN-RNN approach for classifying fake and real news articles. In the first part, CNN has been used for capturing local text patterns and understanding the contextual n-grams that are present in the text.

Here, the CNN model used in this research is a 1D CNN which is designed specifically for text data. Also, the 1D-CNN is able to handle sequential input data, such as word embeddings or text sequences. ReLU activation function has been applied. Dropout Layers have been introduced for regularization to prevent overfitting. In the flatten layer, 3D layers is converted into 1D vector where the last layer is used for the classification. The description of the various parameters used in the CNN have been described in the Table 1.

Table1: Description of the various parameters used in the CNN model

Layer Type	Layer Name	Hyperparameters/ Configuration	Output Dimensions	Description/ Role
Input Layer	Input	Input Shape: (sequence_length, embedding_dim)	(batch_size, 100, 300)	Takes the preprocessed text data as a sequence of word embeddings.
		Example: (100, 300)		
Convolutional 1D	Conv1D_Layer1	Filters: 128	(batch_size, 96, 128)	Applies 128 filters of size 5 to the input, capturing n-gram features and producing 128 feature maps.
		Kernel 5		
		Stride: 1		
		ReLU		
Pooling 1D	MaxPooling1D_Layer1	Pool Size: 2	(batch_size, 48, 128)	Reduces the dimensionality for window of size 2.
		Stride: 2		
Dropout	Dropout_Layer1	Dropout Rate: 0.5	(batch_size, 48, 128)	Randomly sets 50% of the units to zero during each forward pass to prevent overfitting.
Convolutional 1D	Conv1D_Layer2	Filters: 64	(batch_size, 46, 64)	Applies 64 filters of size 3 to capture finer local features and phrases in the text.
		Kernel 3		
		Stride: 1		
		ReLU		
Pooling 1D	MaxPooling1D_Layer2	Pool Size: 2	(batch_size, 23, 64)	Further reduces dimensionality, retaining only the most prominent features from the previous layer.
		Stride: 2		
Dropout	Dropout_Layer2	Dropout Rate: 0.5	(batch_size, 23, 64)	Additional dropout for regularization and improved generalization.
Flatten	Flatten_Layer	N/A	(batch_size, 1472)	Converts the 3D into a 1D vector
Dense (Fully Connected)	Dense_Layer1	Units: 128	(batch_size, 128)	Fully connected layer that combines learned features for final classification.
		Activation: ReLU		
Output Layer	Output	Units: 1	(batch_size, 1)	Produces the final probability for binary classification (fake or real news).
		Activation: Sigmoid		

3.2 Implementation of the RNN architecture

The Recurrent Neural Network (RNN) component in this model architecture plays a pivotal role in capturing the temporal dependencies and contextual flow of the text data. Given the sequential nature of text, RNNs, and LSTM network are most trustable models where the long-term dependencies between words and phrases in a news article. LSTM layers are a specialized type of RNN where the vanishing gradient problem has been removed by incorporating a gating mechanism that controls the flow of information through the network. Each LSTM cell consists of three main components:

1. Forget Gate (f_t) : Determines the information should be discarded from the cell state based on the previous hidden state and current input which is shown mathematically:.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

where h_{t-1} : Previous hidden state. x_t current input W_f ,

2. Input Gate i_t : Decides which new information will be added to the cell state which is represented by mathematically:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

The candidate values for the cell state are computed using the tanh function:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (3)$$

3. Output Gate o_t : Determines what information will be output from the cell which is represented as mathematically

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

In the LSTM Configuration, each LSTM layer in the proposed model uses 64 or 128 units (neurons). These units capture sequential dependencies and are capable of modeling relationships between words that are far apart in the text sequence. The LSTM layer receives a 2D tensor from the CNN component. Each input sequence has been reduced to a fixed size. After processing, the LSTM outputs a hidden state which captures the contextual meaning of words in the given sequence.

3.3 Implementation of the Hybrid CNN-RNN Architecture

The hybrid CNN-RNN architecture combines the strengths of both CNNs and RNNs to provide a comprehensive solution for fake news detection. Table 2 describe the architectural components.

Table 2 : Architecture of Hybrid CNN-RNN architecture

Step	Component	Description	Input Shape	Output Shape	Role/Function
1	CNN Layers	1D filters for extracting the n-grams .	(batch_size, sequence_length, embedding_dim)	(batch_size, reduced_length, num_filters)	Captures local dependencies and n-gram features.
2	Reshaping	Reshapes the CNN output into a format suitable for LSTM input.	(batch_size, reduced_length, num_filters)	(batch_size, sequence_length, num_features)	Each filter is treated as a separate feature vector for each time step.
3	LSTM Layers	Sequentially processes the CNN features to capture long-term dependencies and contextual meanings.	(batch_size, sequence_length, num_features)	(batch_size, hidden_units)	Models sequential dependencies and context across text sequences.
4	Fully Connected Layer	Dense layer that combines the contextual features from LSTM.	(batch_size, hidden_units)	(batch_size, 1)	Combines learned features for final classification.
5	Output Layer	Applies sigmoid activation to produce the final probability of fake or real news.	(batch_size, 1)	(batch_size, 1)	Produces the probability for binary classification (fake or real).

4. Selection of approximate model

The selection of the appropriate model lies between the different parameters while calculating accuracy. Here, CNN and RNN models are employed in order to classify fake and real image texts. CNN operates on grid-like data and this kind of model detects information at the smallest block atomically and hierarchically by using convolution and pooling layers. Along with that, one-dimensional neural networks can provide successful 1-D text classification useful to elicit sequential local patterns and features inherent in the news headlines. The CNNs adopted a neural network that pretended to resemble the way visual data is interpreted by the visual cortex of the brain. With the overlay of neural convolutional filters upon input sequences, CNNs would have characters of its own that they could use in text classifications such as n-grams and local semantics. Also, local semantics would be one thing that would contribute to the process. Regarding the practical solutions, CNN is well-known for its input-computational efficiency and powerful image-recognition abilities, and it works perfectly for both image and text processing, even speech recognition. The model takes advantage of CNNs which allows it to learn directly hence the features that are important to text classification are captured and processed by the model which at the end of it all may predict based on certain patterns from the present text.

The architecture of RNNs are applied to model sequential or time-ordered datasets, is an example of a neural network type that can be used in this way. The feedforward neural networks do not preserve or change the hidden states after the sending of the signals from previous time steps. Scenarios of linguistic context or long-term and transitional dependencies can efficiently be processed by RNN thus resulting in potential use cases including natural language processing models and sentiment analysis as areas of RNN true power. RNNs have brought a very big impact and have been driving the evolution of the industry initiatives which is about different fields involving sequence data like NLP, SR, as well as machine translation. Text classification refers to a type of constructive where a continuous output is dependent upon a condition in which a sequence is assigned representations for its feedback and later interpretation and prediction are made if true.

To be successful in combining them, the features of both reading comprehension models can be integrated so that the negative effects of each text process on its nature can be restored. Our model includes 1D-CNN layers for obtaining the local features from the headings and LSTM layers to model the dependency on time where the headings are a sequence which is performed through the sequence-to-sequence mapping. This hybrid approach enables the model to benefit from the capabilities of both architectures: the fact that the CNNs are provided with the capability of detecting fine-grained details of locations and act as an enabler of RNNs to incorporate context and sequential recurrence opens the opportunity to model dependencies between elements. Thus, by integrating two types of networks: Here the model incorporates the network of convo and recurrent neural networks which are famously recognized for their classification functions of the content of any article so that they present authentic output whenever the textual aspects of such articles are being considered.

5. Simulation Results and Discussions

Data preprocessing was the first step in the simulation process, which aimed at cleaning and standardizing the ‘SherLockFakenewsNetOriginal.csv’ dataset. Among other techniques that were used to improve data quality include HTML tag removal, URL elimination, and noise reduction. Feature engineering encompassed tokenization, padding and incorporation of GloVe word embeddings for capturing semantic relationships. The hybrid model was fed by a split dataset where local features were extracted by CNN layers as well as capturing sequential dependencies by RNN layers. The performance of the model was assessed using several metrics which were then visualized through confusion matrices and ROC curves. Simulations showed that this model could effectively distinguish between fake news articles from real ones.

5.1 Experimental setup

This section provides an overview of the computational resources, software environment, and training configuration used for implementing and evaluating the proposed hybrid CNN-RNN model. The experimental setup was carefully designed to ensure efficient training, robust evaluation, and reproducibility of results. The experiments were conducted on a high-performance computing environment to leverage the computational power needed for deep

learning model training. The hardware Specifications of Personal Computer (PC) with GPU of Nvidia RTX 2080 Ti with 11 GB GDDR6 VRAM, RAM of 64 GB DDR4, and 1 TB NVMe SSD. The implementation of the hybrid CNN-RNN model was carried out on Python 3.7, TensorFlow 2.5, Keras 2.4. For the data Manipulation and analysis, Pandas 1.2.4 and NumPy 1.19.5 has been used. For the Visualization Tools, matplotlib 3.4.2 and Seaborn 0.11.2 version has been applied. Table3 describes the hyperparameters of the proposed model

Table3: Description of the hyperparameters of the proposed model

Hyperparameter	Value/Configuration	Description
Learning Rate	0.001	Controls the step size. A lower value ensures smooth convergence.
Batch Size	128	Number of samples processed. Balances computational efficiency and convergence speed.
Number of Epochs	50	Number of complete passes through the training dataset. Early stopping is used if validation loss does not improve for 5 epochs.
Dropout Rate	0.5	Prevents overfitting
Optimizer	Adam	An optimizer that adapts the learning rate and uses momentum, accelerating convergence and enhancing stability.
Loss Function	Binary Cross-Entropy	Computes the value of the predicted and actual binary labels (fake or real news). Ideal for binary classification tasks.
Activation Function	ReLU (Hidden Layers)	Introduces non-linearity in intermediate layers, allowing the network to capture complex patterns and relationships.
	Sigmoid (Output Layer)	Outputs probability scores for binary classification, producing values between 0 and 1 for each class.

Upon evaluation, the model achieved an accuracy of approximately 96.18% on the training data. The high accuracy of 96.18% on the training data indicates that the model has effectively learned from the training dataset. The capability of the model has been simulated for segregating the news articles that are used for training by considering the relevant patterns and features. The accuracy of 91.80% on the testing data reflects the model's generalization ability to unseen data. Despite not being exposed to the testing dataset during training, the model performs well in classifying news articles, indicating that it can effectively generalize its learned knowledge to new and unseen instances as shown in Fig. 4.

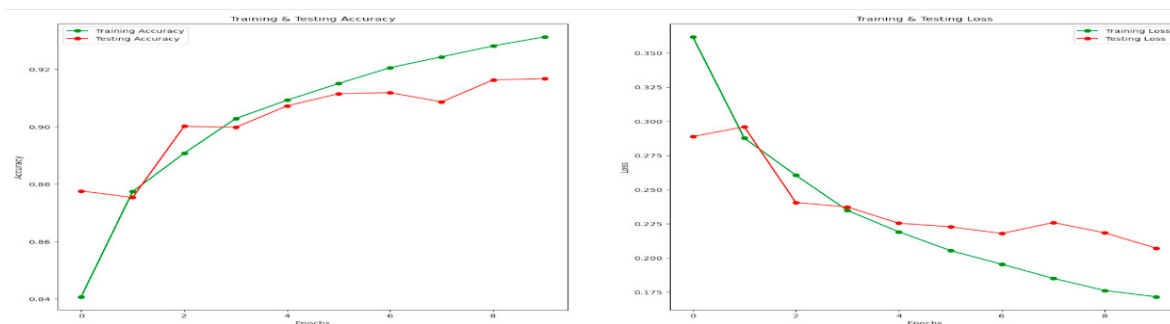
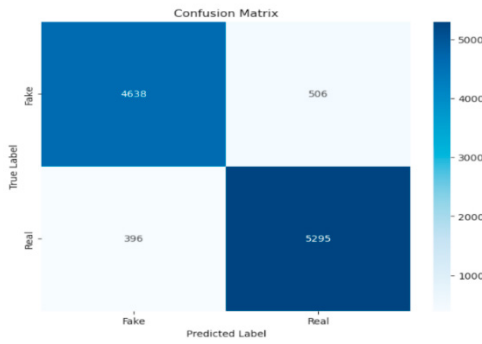


Fig. 4. Summary of histograms for fake and real data

Fig. 4 illustrates the performance of the proposed model over multiple epochs. It consists of two subplots: the left subplot shows how the model's accuracy changes, while the right side observes the changes in the loss. Here, it has been observed that the profession of the line and convergence behaviour. The gradual increase in training and testing accuracy, along with the decrease in loss, indicates that the model is learning effectively without significant

overfitting.

The confusion matrix as given in the Fig.5 is an essential tool for assessing the performance of a classification model by providing a detailed breakdown of predictions. It clearly illustrates the number of true positives, true negatives, false positives, and false negatives. However, to enhance the clarity of Figure 5, the axes should be labeled appropriately to indicate which rows and columns represent "Fake News" and "Real News" predictions and true labels, respectively.



(a)

	precision	recall	f1-score	support
Fake	0.92	0.90	0.91	5144
Not Fake	0.91	0.93	0.92	5691
accuracy			0.92	10835
macro avg	0.92	0.92	0.92	10835
weighted avg	0.92	0.92	0.92	10835

(b)

Fig. 5 (a) Confusion matrix for the proposed CNN-RNN model (b) Different parameters of the confusion matrix

Precision computes the TP predictions out of all instances which are given mathematically as

$$\text{Precision} = \frac{TP}{TP+FP} \quad (5)$$

Recall measures proportion of TP predictions out of all actual positive instances as given.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (6)$$

Accuracy is given as

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} \quad (7)$$

ROC curve is a fundamental tool used to evaluate binary classifiers' ability to diagnose. Originating from signal detection theory, the ROC curve presents a true positive rate (sensitivity) against a false positive rate (1-specificity), at different threshold settings. This kind of visualization helps researchers explore the trade-off between sensitivity and specificity as given in Fig.6.

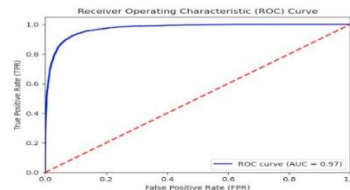


Fig.6 ROC Curve for CNN-RNN hybrid model

Fig.7 presents the ROC curve, articles which suggests that the model performs well in differentiating between the two classes. This demonstrates the model's high diagnostic capability and confirms its effectiveness in practical fake news detection scenarios. The proposed model has been compared with recent works as shown in the Table 4. From the table, it has been observed that the classification accuracy, Precision, Recall and F1 scores of the proposed hybrid model has achieved at a higher rate due to the strengths of CNN and RNN model.

Table 4: Comparison of related works with the proposed hybrid model

Model	Algorithm	Dataset	Accuracy	Precision	Recall	F1-Score
Proposed Hybrid Model	CNN + RNN	SherLockFakenewsNetOriginal.csv	91.80%	0.92	0.9	0.91
Divya et al. (2024) [19]	LSTM-CNN Hybrid	FakeNewsNet	89.70%	0.89	0.88	0.88
Mahalat et al. (2024) [20]	Random Forest	Kaggle Fake News Dataset	83.40%	0.85	0.81	0.83
Nawaz et al. (2024) [21]	Decision Trees	BuzzFeed News	79.30%	0.78	0.75	0.76
Alnabhan. (2024) [22]	BERT	FakenewsNet & ISOT	90.20%	0.91	0.88	0.89
Goldani et al. (2020) [23]	Deep Neural Network	ISOT	87.40%	0.88	0.86	0.87

6. Conclusion

In this paper, the research demonstrates that machine learning models, specifically a hybrid model, which combines CNNs and RNNs can be effectively used to identify fake news. The hybrid model achieved impressive performance of testing accuracy of 91.80% by leveraging Kaggle's rich dataset and using meticulous data preprocessing and feature engineering. Key contributions of this study include comprehensive pre-processing of textual data such as removing HTML tags, removing URLs, noise removal, tokenization, padding, and integration of GloVe word embeddings. Doing this helped ensure that the data fed into the model is clean and well-structured for it to understand meaningful patterns and semantic relationships among news headlines. This study's findings reveal that CNNs capture local textual patterns while RNNs catch up on sequential dependencies. Furthermore, precision, recall, and F1-score metrics are very high implying that the model is robust in distinguishing between real news articles from fake ones. In addition, ROC curve analysis supported these findings making it practical in real-life circumstances.

Although the proposed model has performed well, however there are some limitations and challenges to ensure its broader applicability and ethical use. One key issue is that the dataset might not fully capture all the diverse forms of misinformation, which could cause the model to struggle when encountering new or unfamiliar types of fake news. Additionally, imbalanced data could lead to biased predictions, making it more likely to favor the class that has more representation. Ethical concerns also arise, as solely depending on automated detection could unintentionally result in censoring legitimate content or unfairly labeling truthful articles as fake. To address these issues, future work should focus on expanding the dataset to include a wider variety of topics, continuously updating the data to keep up with new trends, and incorporating human supervision to ensure the system is transparent and accountable. Taking these steps would improve the model's reliability, fairness, and readiness for real-world use.

References

- [1] Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection on social media: A data mining perspective. *ACM SIGKDD explorations newsletter*, 19(1), 22-36.
- [2] Filip Jouhar, J., Pratap, A., Tijo, N., & Mony, M. (2024). Fake News Detection using Python and Machine Learning. *Procedia Computer Science*, 233, 763-771.
- [3] Mahalat, M. H., Priyadarshini, S. B. B., Swain, S., Sahoo, S., Mohapatra, A., & Das, M. (2024). A Comparative Study of Various Machine Learning (ML) Approaches for Fake News Detection in Web-based Applications. In *Microbial Data Intelligence and Computational Techniques for Sustainable Computing* (pp. 59-76). Singapore: Springer Nature Singapore.

- [4] Dev, D. G., Bhatnagar, V., Bhati, B. S., Gupta, M., & Nanthaamornphong, A. (2024). LSTMCNN: A hybrid machine learning model to unmask fake news. *Heliyon*, 10(3).
- [5] Dheyaa Radhi, A., Ali Hussein Al Naffakh, H., Fuqdan, A. I., A Hakim, B., & Al-Attar, B. (2024). A comprehensive review of machine learning-based models for fake news detection. In *BIO Web of Conferences* (Vol. 97, p. 00123). EDP Sciences.
- [6] Su, Q., Wan, M., Liu, X., & Huang, C. R. (2020). Motivations, methods and metrics of misinformation detection: an NLP perspective. *Natural Language Processing Research*, 1(1-2), 1-13.
- [7] Singh, Manjari, Subhranil Das, and Sudhansu Kumar Mishra (2020). "Static obstacles avoidance in autonomous ground vehicle using fuzzy logic controller." In *2020 International Conference for Emerging Technology (INCET)*, pp. 1-6.
- [8] Mishra, Sudhansu Kumar, and Subhranil Das (2019). "A review on vision based control of autonomous vehicles using artificial intelligence techniques." In *2019 International Conference on Information Technology (ICIT)*, pp. 500-504. IEEE.
- [9] Kumari, Rashmi, Shivani Goel, and Subhranil Das (2023). "Mathematical modelling of dendritic complexity mechanism in Alzheimer's disease." In *AIP Conference Proceedings*, vol. 2872, no. 1. AIP Publishing.
- [10] Dadhich, Arushi, Subhranil Das, and Raghwendra Kishore Singh (2024). "Empirical Evaluation of Deep Learning Architectures in the Early Detection of Alzheimer's Disease through MRI Data Analysis." In *Proceedings of International Conference on Intelligent Systems and New Applications*, (2): 27-31.
- [11] Ahmad, Tanzeel, Arnav Talaan, Vishal Gola, Subhranil Das, and Raghwendra Kishore Singh (2024). "Advancing Early Diagnostic Accuracy for Alzheimer's Disease Through the Integration of Machine and Deep Learning Paradigms by Applying Multisource Datasets." In *Proceedings of International Conference on Intelligent Systems and New Applications*, 2:42-46.
- [12] Kumari, Rashmi, Shivani Goel, and Subhranil Das. (2022). A 3D Convolutional Neural Network Approach for Diagnosing Alzheimer's Disease using Modified Owl Search Optimization Technique." In *TENCON 2022-2022 IEEE Region 10 Conference (TENCON)*, pp. 1-7. IEEE.
- [13] Kumari, Rashmi, Shivani Goel, and Subhranil Das. (2022). Using SVM for Alzheimer's Disease detection from 3D T1MRI. In *2022 IEEE 21st Mediterranean Electrotechnical Conference (MELECON)*, pp. 600-604.
- [14] Das, Subhranil, and Rashmi Kumari. (2021) "Application of Extended Hough Transform Technique for Stationary Images in Vehicle License Plate." In *2021 6th International Conference for Convergence in Technology (I2CT)*, pp. 1-4. IEEE, 2021.
- [15] Oh, Jeong-su. (2021). Performance Analysis of Hough Transform Using Extended Lookup Table. *Journal of the Korea Institute of Information and Communication Engineering* 25,12:1868-1873.
- [16] Pennycook, Gordon, and David G. Rand (2021). The psychology of fake news." *Trends in cognitive sciences* 25, no. 5: 388-402.
- [17] Zhang, Xichen, and Ali A. Ghorbani. (2020) An overview of online fake news: Characterization, detection, and discussion." *Information Processing & Management* 57, 2:102025.
- [18] Zhou, Xinyi, and Reza Zafarani. (2020) A survey of fake news: Fundamental theories, detection methods, and opportunities. *ACM Computing Surveys (CSUR)* 53, no. 5: 1-40.
- [19] Divya, T. V., & Mohanty, F. (2024). An Adaptive Word Vector Integrated With Enhanced Lstm For Classification Of Fake News. *Journal of Electrical Systems*, 20(7s), 4062-4071.
- [20] Mahalat, M. H., Priyadarshini, S. B. B., Swain, S., Sahoo, S., Mohapatra, A., & Das, M. (2024). A Comparative Study of Various Machine Learning (ML) Approaches for Fake News Detection in Web-based Applications. In *Microbial Data Intelligence and Computational Techniques for Sustainable Computing* (pp. 59-76). Singapore: Springer Nature Singapore.
- [21] Nawaz, M. Z., Nawaz, M. S., Fournier-Viger, P., & He, Y. (2024). Analysis and Classification of Fake News Using Sequential Pattern Mining. *Big Data Mining and Analytics*, 7(3), 942-963.
- [22] Alnabhan, Mohammad Q., and Paula Branco. "Evaluating Deep Learning for Cross-Domains Fake News Detection." In *International Symposium on Foundations and Practice of Security*, pp. 40-51. Cham: Springer Nature Switzerland, 2023.
- [23] Goldani, M. H., Safabakhsh, R., & Momtazi, S. (2021). Convolutional neural network with margin loss for fake news detection. *Information Processing & Management*, 58(1), 102418.